ABSTRACT

Topic: Detection of Malicious Instructions in Bulgarian for Large Language Models

This thesis addresses the problem of detecting malicious instructions in Bulgarian directed at large language models (LLMs). As artificial intelligence systems become increasingly integrated into real-world applications, protecting them against attempts to bypass their safety mechanisms has become an important challenge.

As part of this work, a Bulgarian-language dataset containing both safe and malicious instructions was developed. The original data were further expanded through automated text transformations and paraphrasing techniques aimed at generating more diverse and difficult-to-detect malicious inputs.

Several models were evaluated, including *XLM-RoBERTa*, *Mistral-7B-Instruct-v0.3*, *GPT-4o*, and *BgGPT-7B-Instruct-v0.2*. The evaluation included zero-shot and few-shot settings, as well as fine-tuning on the developed dataset. Standard classification metrics, including Accuracy, Precision, Recall, and F1-score, were used to assess model performance.

In addition to the main experiments, the best-performing model was evaluated on an independent dataset from the *HackAPrompt* competition. The results demonstrate that fine-tuning on specialized data significantly improves the ability of language models to recognize malicious instructions. The highest performance was achieved by the fine-tuned *BgGPT-7B-Instruct-v0.2* model, which attained high classification accuracy and correctly identified all malicious instructions in the test set. Error analysis showed that misclassifications were rare and primarily associated with short or contextually ambiguous queries.

The obtained results confirm that fine-tuning on domain-specific safety data is an effective approach for improving the security of large language models in the Bulgarian language context. As part of the thesis, a prototype system was also developed for classifying user instructions into two categories: safe and malicious. Such a system can serve as a foundation for building a protective filtering layer for applications based on large language models.

Keywords: large language models, artificial intelligence, jailbreak, prompt injection, language model security, text classification, BgGPT, XLM-RoBERTa.